# Edward Stevinson

***Research scientist specialising in mechanistic interpretability and adversarial robustness***

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Machine learning researcher and engineer with ten years of industry experience, now pursuing a PhD in AI safety. Current focus is on understanding representation geometry and how it shapes adversarial vulnerability.

## Education

**Imperial College London, PhD.** 2022 - Present

Imperial-X scholarship in Safe AI under Prof. Tolga Birdal & Prof. Alessio Lomuscio. Interdisciplinary approach to robustness spanning mechanistic interpretability and neural network verification.

**University of Edinburgh -** MSc in Artificial Intelligence (MSc) - 1. Distinction 2015 - 2016

Machine learning specialism focusing on probabilistic modelling, deep learning & multi-agent systems.

**Trinity College, Oxford -** MEng in Engineering, Economics and Management - 2.1 2011 - 2015

Specialised in machine learning, robotics and algorithmic game theory.

## Selected Publications

**Stevinson, E,** Prieto, L, Barsbey, M, and Birdal, T, *Adversarial Vulnerability from Interference Between Features in Superposition,* 2026, ICML.

Prieto, L, **Stevinson, E,** Barsbey, M, and Birdal, T, *Correlations in the Data Lead to Semantically Rich Feature Geometry Under Superposition,* 2026, ICLR.

**Stevinson, E*,** Graham, R*, Richter, L*, Chia, A*, Miller, J, and Bloom, J, *ContextBench: Modifying Contexts for Targeted Latent Activation,* 2026, ICLR.

Joseph, S, Hufe, L, **Stevinson, E,** Graham, R, Vadi, Y, Bzdok, D, Sharkey, L, and Richards, B, *Prisma: An Open Source Toolkit for Mechanistic Interpretability in Vision, 2025*, MIV Workshop @ CVPR **(Oral)**

Manginas, V, Manginas, N, **Stevinson, E,** Varghese, S, Katzouris, N, Paliouras, G, and Lomuscio, A, *A Scalable Approach to Probabilistic Neuro-Symbolic Robustness Verification*, 2025, NeSy **(Oral & Outstanding Paper Award)**

**Stevinson, E,** and Lomuscio, A, 2024, Reducing Return Volatility in Neural Network-Based Asset Allocation via Formal Verification and Certified Training, ICAIF **(Oral)**

Hoelscher-Obermaler, J, and **Stevinson, E**, 2022, Leveraging knowledge graphs to update scientific word embeddings using latent semantic imputation, WIESP @ AACL-IJCNLP 2022 **(Oral)**

## Professional Experience

**LASR Labs** - Research Fellowship 02.2025 - 05.2025

- AI safety research focusing on eliciting undesirable behaviours in LLMs by generating fluent modifications to prompts that maximally activate SAE features of interest.

**Safe Intelligence** - Research Scientist 03.2023 - 02.2025

- Lead developer of the neural network verification toolkit, transforming research code into open-source production-grade software that ensures network robustness under input perturbations.
- Enabled verification of time series models, opening up new tasks and customers, by implementing verification for softmax and 1D convolutional layers.

**Napo** - Senior Software Engineer 04.2022 - 09.2022

- Python backend developer for the fast growing insurance startup. As a member of the growth team, we grew the customer base from 1k to 10k over six months.

**Upside Ltd** - Lead Data Scientist and Co-Founder 11.2020 - 02.2022

- Led 3-person team building a workflow analytics platform to empower teams by surfacing key metrics.

**Eigen Technologies** - Solutions Engineer 06.2019 - 11.2020

- AI company that automates the extraction & classification of information from legal documents.

**IBM** - Software Developer 06.2017 - 01.2019

- Transformed i2's law enforcement analysis software from desktop application to web platform.

## Technical Skills

- 5+ years professional ML experience, 7+ years professional coding experience.
- **Research frameworks & libraries:** Extensive experience with PyTorch, Weights & Biases, PyTorch Lightning, NumPy, and Pandas.
- **Neural network interpretability:** Transformers (Hugging Face), TransformerLens, SAELens, Prisma, activation patching, steering, causal tracing.
- **Neural network verification:** Contributed to VeriNet and Venus, and extensively used α,β-CROWN, for verification and certified training.
- **MLOps:** Experience deploying ML solutions using Airflow, Dagster, MLFlow, Docker, Kubernetes, GitHub Actions & Pachyderm.

## Teaching

- Deep Learning. T.A., Imperial College London, 2023.
- Natural Language Processing. T.A., Imperial College London, 2023.
- Demystifying Machine Learning. T.A., Imperial College London, 2023.
- Codebar programming mentor, 2020-2022.

## AI Safety Projects

2025 **LogML**

Project using singular learning theory to predict phase transitions in feature learning during grokking in neural networks.

2024 **ViT-Prisma** *(Github)*

Open source contributor to the Prisma mechanistic interpretability toolkit for multimodal models.

**Apart Research Hackathon - Reprogramming AI Models** *(Link)*

Explored how LLMs respond to adversarial attacks at the feature level. Our second place entry investigated whether latent activations could help identify harmful prompts.

2023 **AI Safety Camp**

Drafting high-risk policy proposals for the EU Safety Act as part of AISC8.

2018 **Towards Fair AI: tackling bias in predictive probabilistic models**

Essay submission for GeneralAI's 'mitigating the risks of the AI race' competition.

2016 **Utilising Policy Types to Achieve Effective Ad Hoc Coordination**

Research on autonomous agents in multi agent systems with no mechanisms for prior coordination.